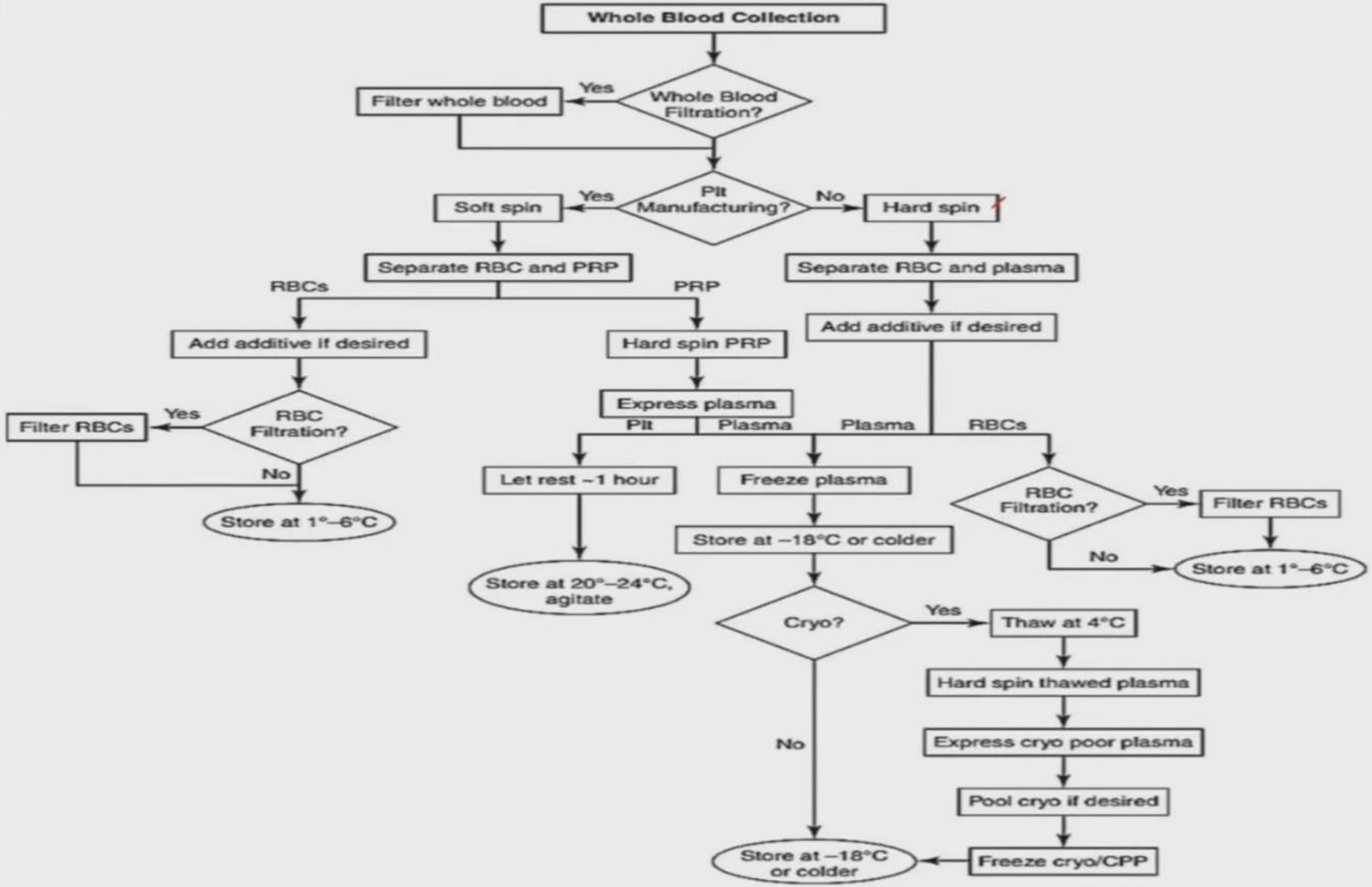
OVERVIEW OF THE PROCESS FOR MANUFACTURING WHOLE BLOOD INTO COMPONENTS

SOUTHWESTERN UNIVERSITY
PHINMA



BLOOD COMPONENTS

SOUTHWESTERN UNIVERSITY

Table 15–2 Blood Component Characteristics

Component	Shelf Life	Storage Temp.	Quality Control	Volume	Dosage
Whole blood (WB)	CPD 21 d CPDA-1 35 d CP2D 21 d	1°–6°C	N/A	450–500 mL	↑ hgb 1 g/dL ↑ hct 3%
Whole blood irradiated	Original expiration or 28 d post-irradiation	1°–6°C	25 Gy to center of canister	450–500 mL	↑ hgb 1 g/dL ↑ hct 3%
RBCs	CPD 21 d CPDA-1 35 d CP2D 21 d ACD 21 d AS 42 d	1°–6°C	No additive: hct ≤ 80% Additive: N/A	250–300 mL	↑ hgb 1 g/dL ↑ hct 3%
RBC aliquots	Closed system: same Open system: 24 hr	1°–6°C		varies	10 mL/kg ↑ hgb 2 g/dL
RBC irradiated	Original outdate or 28 d post-irradiation	1°–6°C	25 Gy to center of canister	250–300 mL	↑ hgb 1 g/dL ↑ hct 3%
RBC leukoreduced	Closed system: same Open system: 24 hr	1°–6°C	$<5 \times 10^6$ WBCs ≥ 85% RBC recovery	250–300 mL	↑ hgb 1 g/dL ↑ hct 3%
Washed RBCs	24 hr	1°–6°C	hct 70–80%	180 mL	↑ hgb 1 g/dL ↑ hct 3%
Frozen RBCs	10 years	≤ –65°C			
RBC deglycerolized	24 hr	1°–6°C	80% RBC recovery <1% glycerol	180 mL	↑ hgb 1 g/dL ↑ hct 3%
Platelets, whole-blood derived (RD)	5–7 d	20°–24°C	$\geq 5.5 \times 10^{10}$ plts pH ≥ 6.2	50–70 mL	↑5k–10k/μL

Table 15–2 Blood Component Characteristics—cont'd

Component	Shelf Life	Storage Temp.	Quality Control	Volume	Dosage
Platelets, apheresis (SD)	5–7 d	20°–24°C	≥ 3×10^{11} plts pH ≥ 6.2	200–400 mL	↑30k–60k/μL
Platelets, irradiated	5 d	20°–24°C	25 Gy to center of canister	Same	Same
Platelets, pooled	4 hr	20°–24°C	pH ≥ 6.2	Varies	Varies
Platelets, leukoreduced	5 d	20°–24°C	pH ≥ 6.2 RD: <8.3×10^5 WBCs SD: <5×10^6 WBCs	RD: 40–70 mL SD: 200–400 mL	RD: ↑ 5–10k/μL SD: ↑ 30–60k/μL
FFP AFFP	1 yr 7 yr 1 yr	≤–18°C ≤–65°C ≤–18°C		200–380 mL 200–1,000 mL	↑Factor 20%–30% 10–20 mL/kg
PF24	1 yr 7 yr	≤–18°C ≤–65°C		200–380 mL	Same
LP	5 days after WB expires	1–6°C liquid		200–380 mL	Not well characterized
Cryoprecipitate	Frozen: 1 yr Thawed: 6 hr Pooled (open): 4 hours	≤–18°C 20°–24°C	FVIII: 80 IU Fibrinogen: 150 mg	10–25 mL	↑Fibrinogen 5–10 mg/dL
Cryo-reduced plasma	1 yr	≤18°C		200–380 mL	
Granulocytes	24 hr	20°–24°C	≥1.0×10^{10}	200–600 mL	1–2 × 10^{10}/infusion four daily doses
Granulocytes, irradiated	24 hr	20°–24°C	≥1.0×10^{10}	200–600 mL	Same

hgb = hemoglobin; hct = hematocrit; plts = platelets; WB = whole blood; SD = single donor; RD = random donor

Table 16–1 Blood Components and Plasma Derivatives

Category	Major Indications	Mode of Action	Risks	Special Considerations
Whole blood	Symptomatic anemia with large-volume deficit	Increases oxygen-carrying capacity Increases blood volume	Infectious diseases Hemolytic, septic/toxic, allergic, febrile reactions Iron overload, TACO, TRALI, TA-GVHD	Must be ABO identical
Red blood cells: RBCs (adenine-saline added), RBC pheresis	Symptomatic anemia; red blood cell exchange transfusion	Increases oxygen-carrying capacity	Same as whole blood category, above	Must be ABO compatible
Red blood cells: deglycerolized, washed	IgA deficiency with anaphylactoid/anaphylactic reaction Severe allergic reactions, rare donors, symptomatic anemia; red blood cell exchange transfusion	Deglycerolization removes plasma proteins. Risk of allergic and febrile reactions reduced; increases oxygen-carrying capacity	Same as RBCs above Hemolysis due to incomplete deglycerolization can occur.	Must be ABO compatible
Red blood cells: leukocyte-reduced	Symptomatic anemia Febrile reactions due to leukocyte antibodies Reduction of CMV transmission and HLA alloimmunization	Leukocyte reduction is achieved by filtration: (1) soon after collection (prestorage) or (2) after varying periods of storage in the laboratory.	Same as RBCs Hypotensive reaction may occur if bedside leukocyte-reduction filter is used.	RBCs leukocyte-reduced must have a residual content of leukocytes $<5 \times 10^6$ WBC and ≥85% of the original red blood cell content.

Platelets/apheresis platelets	Bleeding due to thrombocytopenia or platelet function abnormality including antiplatelet drugs Prevention of bleeding from marrow hypoplasia	Improves hemostasis Apheresis platelets may be HLA (or other antigen) selected.	Infectious diseases Septic/toxic, allergic, febrile reactions TACO, TRALI,- TA-GVHD	One unit of platelets derived from a whole blood collection usually contains $\geq 5.5 \times 10^{10}$ platelets suspended in 40 to 70 mL of plasma. One unit of apheresis platelets usually contains $\geq 3 \times 10^{11}$ platelets/unit and is the therapeutic equivalent of 4 to 6 units of platelets.
Platelets leukocytes reduced/apheresis platelets leukocytes reduced	Same as platelets above Prevention of febrile reactions, HLA alloimmunization and CMV infection	Same as platelets above	Same as platelets above	May be prepared using an open or closed system. One unit of platelets leukocyte-reduced should contain $>5.5 \times 10^{10}$ platelets and $<8.3 \times 10^{5}$ leukocytes. Components prepared using an open system will expire 4 hours after preparation.
Apheresis granulocytes	Neutropenia with infection unresponsive to appropriate antibiotics	Increases the level of granulocytes to phagocytize and kill bacterial and fungal infections	Infectious diseases Hemolytic, allergic, febrile reactions TACO, TRALI, TA-GVHD	Must be ABO compatible Contain numerous leukocytes and platelets as well as 20 to 50 mL of RBCs; the number of granulocytes in each concentrate is usually $>1 \times 10^{10}$ granulocytes/unit.
Plasma (liquid)	Initial treatment of patients Undergoing massive transfusion	Coagulation support for life-threatening trauma, hemorrhages	Infectious diseases Allergic and febrile reactions, TACO, TRALI	Must be ABO compatible

Table 16–1 Blood Components and Plasma Derivatives—cont'd

Category	Major Indications	Mode of Action	Risks	Special Considerations
Cryoprecipitate pooled/ cryoprecipitated AHF	Hypofibrinogenemia Factor XIII deficiency (Second line therapy for von Willebrand's disease, hemophilia A, and uremic bleeding)	Provides fibrinogen (factor I), factors VIII, XIII, and vWF	Infectious diseases Allergic and febrile reactions	Must contain >150 mg of fibrinogen in each unit of cryoprecipitate Cryoprecipitated AHF should contain >80 IU of factor VIII in each unit.
Fresh-frozen plasma (FFP)	Clinically significant plasma protein deficiencies when no specific coagulation factor concentrates are available TTP	Source of all coagulation proteins and plasma proteins	Infectious diseases Allergic and febrile reactions, TACO, TRALI	Must be ABO compatible
PF24 – (Plasma frozen within 24 hours after phlebotomy)	Same as FFP above	Source of nonlabile plasma proteins Levels of factor VIII are significantly reduced and levels of factor V and other labile plasma proteins are variable compared with FFP.	Same as FFP above	Must be ABO compatible

PF24RT24 (Plasma frozen within 24 hours after phlebotomy held at room temperature up to 24 hours after phlebotomy)	Clinically significant deficiency of stable coagulation factors when no specific coagulation factor concentrates are available TTP	Source of nonlabile plasma proteins Levels of factor V, factor VIII, and protein S are significantly reduced, and levels of other labile plasma proteins are variable compared with FFP.	Same as FFP	Must be ABO compatible
Thawed plasma	Clinically significant deficiency of stable coagulation factors when no specific coagulation factor concentrates are available TTP	Source of plasma proteins and contains stable coagulation factors (factor II, fibrinogen), similar clinically to the levels found in FFP; other factors are variable, being reduced in levels that change over time.	Same as FFP	Must be ABO compatible
Thawed plasma, cryoprecipitate reduced	TTP	Plasma protein replacement for plasma exchange in TTP; this product contains variable levels of albumin, ADAMTS13, and factors II, V, VII, IX, X, and XI.	Same as FFP	Must be ABO compatible Product is deficient in factors I (fibrinogen), VIII, XIII, as well as vWF, cryoglobulin, and fibronectin.

CMV-negative components should be administered to recipients who are:

- CMV-negative and at risk for severe sequelae of CMV infections
- CMV-seronegative pregnant women (mainly for the benefit of fetus)
- CMV-seronegative bone marrow transplants recipients
- CMV-seronegative premature infants.

HEMAPHERESIS/APHERESIS

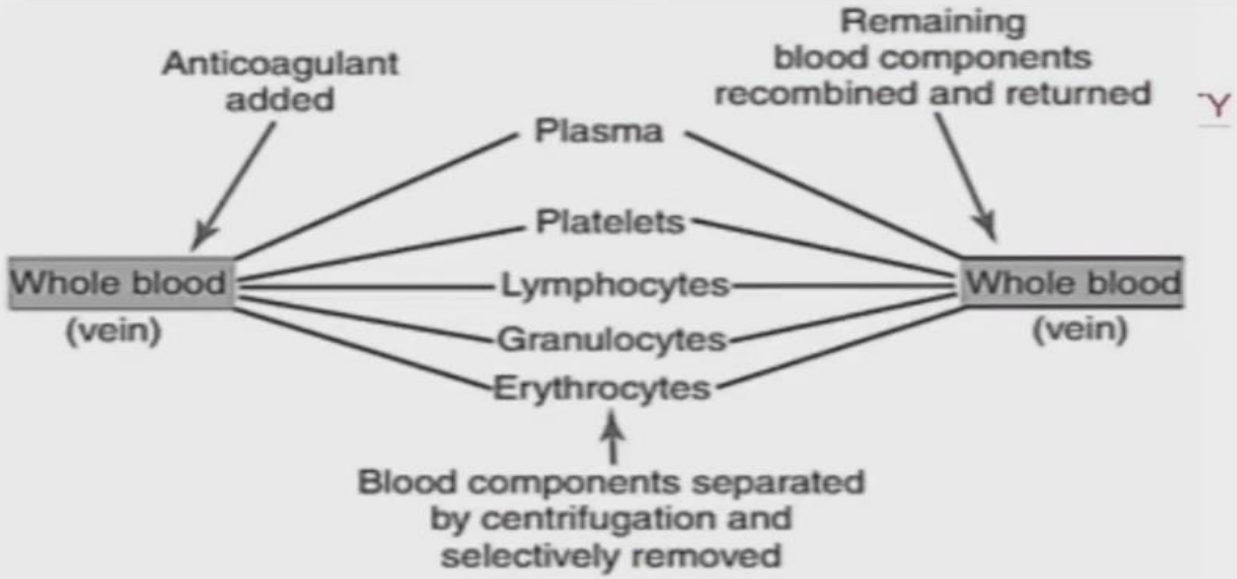
SOUTHWESTERN UNIVERSITY
PHINMA

- Type of blood donation where WB is withdrawn eather from a donor or patient wherein after removal, separation and retention of the desired cellular element or plasma occurs and the remaining products are recombined and returned to the donor or patient
- INTERMITTENT-FLOW CENTRIFUGATION (IFC)
  - ➢ Requires only one venipuncture, in which blood is withdrawn and reinfused through the same needle.
- CONTINUOUS-FLOW CENTRIFUGATION (CFC)
  - ➢ Procedures withdraw, process, and return the blood to the individual simultaneously.
- MEMBRANE FILTRATION
  - ➢ Membrane separators are typically composed of bundles of hollow fibers or flat plate membranes with specific pore sizes

Variables that are considered during an apheresis procedure include:

1. Centrifuge speed and diameter
2. Duration of dwell time of the blood in the centrifuge
3. Type of solutions added, such as anticoagulants or sedimenting agents
4. Cellular content or plasma volume of the patient or donor

HEMAPHERESIS PRINCIPLE:



DONATION FREQUENCY	
APHERESIS COMPONENT COLLECTED	FREQUENCY OF DONATION
2 RBC	16 weeks
Plasma (frequent)	Every 2 days (no more than two times in 7 days)
Plasma (infrequent)	Every 4 weeks (no more than two times in 13 times/year)
Platelets, single apheresis unit	Every 2 days (no more than two times in 7 days,; no more than 24 times in 12 months))
Platelets, doub;e or triple apheresis unit	Every 7 days
Granulocytes	Every 2 days

SOUTHWESTERN UNIVERSITY
PHINMA

1. PLASMAPHERESIS

- The removal and retention of the plasma, with return of all cellular components to the patient.
- This has become synonymous with Plasma Exchange.
- The purpose is to remove the offending agent in the plasma causing clinical symptoms which includes the following:
  - Immune complexes (e.g. SLE)
  - Autoantibodies or alloantibodies (e.g. factor VIII inhibitors)
  - Antibodies causing hyperviscosity (e.g. Waldenstrom's macroglobulinemia)
  - Inflammatory mediators (e.g. fibrinogen and complement)
  - Antibody blocking the normal function of the immune system.
  - Protein-bound toxins (e.g. barbiturate poisoning)
  - Lipoproteins
  - Platelet-aggregating factors
- Donors must have at least **6.0 g/dL total protein**

2. PLATELETPHERESIS

SOUTHWESTERN UNIVERSITY PHINMA

- Donors should have a platelet count of atleast 150, 000 per uL
- Aspirin ingestion within 2 days of donation is cause of temporary deferral

3. LEUKAPHERESIS

- Donors must have an absolute granulocyte count of at least **4 X 10°/L**
- HES (Hydroxyethyl Starch)
  - RBC sedimenting agent used for granulocyte collection which causes red cells to form **rouleaux** thus allowing WBC's to be harvested more efficiently
- Corticosteroids
  - Administered to the donors **12–24 hrs before pheresis** to increase the number of circulating granulocytes by pulling them from the marginal pool.
- Used to treat patients with **leukemia (WBC > 100,000/uL) such as Hairy Cell Leukemia, AML, cutaneous T-cell Lymphoma.**

4. LYMPHOCYTOPHERESIS

- Means of producing immunosuppression in conditions like RA, SLE, Kidney transplant rejection and autoimmune and alloimmune dieases.

5. NEOCYTAPHERESIS

SOUTHWESTERN UNIVERSITY PHINMA

- Transfusion in young RBC's (neocytes) for patients with certain hematologic disorders especially Thalassemia

6. ERYTHROCYTAPHERESIS

- Considered an **exchange procedure**
- Predetermined quantity of red cells is removed from the patient and replaced with homologous blood.
- Used to treat various complications of sickle cell disease, such as priapism and impending stroke.

POSSIBLE ADVERSE REACTION

1. Citrate toxicity
2. Vascular access complications
3. Vasovagal reaction
4. Allergic reactions
5. Hemolysis
6. Circulatory and Respiratory Distress
7. Transfusion-transmitted dieases